First-Principles Derivation of the Steiner Sentence Length Distribution

Speculative Results Series, Paper 67

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Abstract

We prove that Steiner sentences in the Syracuse graph have length distribution $P(\text{sentence length} = k) = 3^{k-1}/4^k$ for all $k \geq 1$. The proof proceeds in three steps. First, we establish the exact Syracuse transition matrix on residue classes modulo 8 from arithmetic alone: the mod-8 map is uniform from classes 1 and 5, while classes 3 and 7 have restricted (forbidden) transitions determined by mod-16 arithmetic. Second, we identify the invariant $d_3 = d_7$ — equal weight on classes 3 and 7 in the surviving (non-terminated) distribution — and show it is preserved by the transition matrix unconditionally, for any value of d_1. Third, we show that the invariant $d_3 = d_7$ is precisely the algebraic condition under which the surviving mass decays by exactly 3/4 per step, giving $P(k) = 3^{k-1}/4^k$.

The proof requires no ergodic theory and no appeal to the Collatz conjecture. It is a direct consequence of two classical arithmetic facts: that $3n + 1 \equiv 2 \pmod 4$ for $n \equiv 3$ or $7 \pmod 8$ (mod-8 forbidden transitions), and that multiplication by 3 is a bijection on $\mathbb{Z}/2^j\mathbb{Z}$ (uniform distribution from classes 1 and 5).

Appendix A provides a complete empirical validation: 10^5 sentences sampled uniformly over $[1, 10^{15}]$, the sentence-start distribution, the within-sentence word entry distribution, the sentence length table comparing the naive $(1/2)^k$ prediction and the theoretical $3^{k-1}/4^k$, and χ^2 goodness-of-fit tests. The naive prediction is rejected with $\chi^2 \approx 10^6$; the theoretical formula is consistent ($p \approx 0.61$).

Status. This paper contains proved theorems and complete empirical validation. All theoretical results are unconditional.

Contents

1 Background and Notation

We work throughout with odd positive integers.

Definition 1.1 (Syracuse map)**.** The *Syracuse map* $S : \mathbb{O} \to \mathbb{O}$ on odd positive integers is

$$S(n) \;=\; \frac{3n+1}{2^{v_2(3n+1)}},$$

where $v_2(m)$ denotes the 2-adic valuation of m.

Definition 1.2 (Steiner sentence)**.** A *Steiner sentence* is a maximal sequence of consecutive applications of S from an initial node $b_0 = S(n)$ with $n \equiv 5 \pmod 8$, terminating at the first node $b_k \equiv 5 \pmod 8$. The *sentence length* is the number of steps k.

Remark 1.3. Steiner sentences are the fundamental composite building blocks of Collatz paths; see Paper 33 [1] for the full Steiner circuit framework. The sentence start $b_0 = S(n)$ is the node immediately following a $5 \pmod 8$ node in the orbit, and the sentence ends at the next $5 \pmod 8$ node.

Remark 1.4 (Why Steiner sentences are the right scale)*.* The Steiner sentence decomposition is chosen precisely so that class $5 \pmod 8$ is *always a sink and never a source* within a sentence. This has a crucial arithmetic consequence.

For $n \equiv 1, 3, 7 \pmod 8$, the 2-adic valuation $v_2(3n+1)$ is a *constant* determined entirely by n mod 8: it equals 2, 1, and 1 respectively (Paper 66 [3], Theorem 2.1). The Syracuse image $S(n)$ mod 8 is therefore completely determined by n mod 8 alone, with exact rational transition probabilities.

For $n \equiv 5 \pmod 8$, by contrast, $v_2(3n + 1) = 3 + v_2(3k + 2)$ where $n = 8k + 5$, and $v_2(3k+2)$ is an unbounded random variable whose expectation requires summing a geometric series (Paper 66). The transition probabilities out of class 5 are still exactly 1/4 each, but they arise from a distribution over infinitely many valuation values, not a single fixed shift.

By restricting to the *interior* of a sentence — the steps from $b_0 \not\equiv 5$ to the terminal word — class 5 never appears as a source. The surviving chain operates entirely on $\{1, 3, 7\}$, where every valuation is a constant and every transition probability is an exact rational. The variable-valuation complexity of class 5 is absorbed at the sentence boundary and plays no role in the counting argument. This is why the sentence length distribution $3^{k-1}/4^k$ admits a clean closed-form proof: at the scale of Steiner sentences, the arithmetic is exact.

Paper 65 [2] was the first attempt at this subject. It observed empirically that $P(\text{sentence length} = k) = \frac{1}{3}(3/4)^k$, equivalently $3^{k-1}/4^k$, and stated the result as a conjecture. The present paper supersedes Paper 65 by providing a complete proof, and incorporates its experimental record in Appendix A.

2 The Syracuse Transition Matrix mod 8

We model the sentence as a Markov chain on residue classes modulo 8. The transition probability $P[a][b]$ is the natural density of odd integers $n \equiv a \pmod 8$ for which $S(n) \equiv b \pmod 8$, i.e. $P[a][b] = \lim_{N\to\infty} \frac{1}{N}\#\{n \le N : n \equiv a, S(n) \equiv b \pmod 8\}$. All results in this paper are unconditional: they require no assumption beyond this standard density definition and do not depend on the Collatz conjecture.

Theorem 2.1 (Syracuse transition matrix mod 8)**.** *The transition matrix on* $\{1, 3, 5, 7\} \pmod 8$ *is:*

$$P \;=\; \begin{array}{c|cccc} & 1 & 3 & 5 & 7 \\ \hline 1 & 1/4 & 1/4 & 1/4 & 1/4 \\ 3 & 1/2 & 0 & 1/2 & 0 \\ 5 & 1/4 & 1/4 & 1/4 & 1/4 \\ 7 & 0 & 1/2 & 0 & 1/2 \end{array}$$

(rows = source class, columns = target class).

Proof. We prove each row by direct arithmetic.

Row 1 (uniform). Write $n = 8k + 1$. Then $3n + 1 = 24k + 4 = 4(6k + 1)$. Since $6k + 1$ is always odd, $v_2(3n + 1) = 2$ exactly and $S(n) = 6k + 1$. Now $6k + 1 \pmod 8$: as k runs through $0, 1, 2, 3 \pmod 4$, we get $6k + 1 \equiv 1, 7, 5, 3 \pmod 8$ respectively. Each residue in $\{1, 3, 5, 7\}$ appears with equal density, so $P[1][b] = 1/4$ for all $b \in \{1, 3, 5, 7\}$.

Row 5 (uniform). Write $n = 8k+5$. Then $3n+1 = 8(3k+2)$, so $v_2(3n+1) = 3+v_2(3k+2)$. Write $3k + 2 = 2^m q$ with q odd; then $S(n) = q$. We need $q \pmod 8$ to be uniform on $\{1, 3, 5, 7\}$.

Since $\gcd(3, 2^j) = 1$ for all $j \ge 1$, the map $k \mapsto 3k + 2$ is a bijection on $\mathbb{Z}/2^j\mathbb{Z}$. As k ranges over any 2^{j+3} consecutive integers, the value $3k + 2$ hits each residue class mod 2^{j+3} exactly 2^j times. Consequently the value $S(8k + 5) \pmod 8$ is equidistributed over $\{1, 3, 5, 7\}$ *in natural density*: the arithmetic core is that, as k ranges over $\{0, 1, \dots, 2^N - 1\}$, exactly 2^{N-j} values satisfy $2^j \mid (3k+2)$ (Paper 66 [3], `uniformity_count`; formalised here as `row5_uniform_dvd`), from which each target residue $b \in \{1, 3, 5, 7\}$ receives limiting density 1/4. Thus $P[5][b] = 1/4$ for all $b \in \{1, 3, 5, 7\}$.

Remark 2.2 (Row 5 is a density, not an exact finite count). Unlike rows 1, 3 and 7 — whose per-residue counts over $\{0, \dots, 2^N - 1\}$ are *exactly* 2^{N-2} or 2^{N-1} (formalised as `count_row1`, `count_row3_*`, `count_row7_*`) — row 5 holds only as a natural-density limit. The finite count $\#\{k < 2^N : S(8k+5) \equiv b \pmod 8\}$ is *not* equal to 2^{N-2} in general: at $N = 6$, $b = 1$ the count is 17, not $16 = 2^{6-2}$. This is machine-verified in the companion Lean development by `row5_count_not_exact` (via `native_decide`). The extra, variable number of halvings $v_2(3n+1) = 3 + v_2(3k+2)$ from class 5 is exactly what prevents an exact finite count while still forcing uniform limiting density.

Row 3 (restricted to $\{1, 5\}$). Write $n = 8k+3$. Then $3n+1 = 24k+10 = 2(12k+5)$. Since $12k$ is even, $12k+5$ is always odd, so $v_2(3n+1) = 1$ exactly and $S(n) = 12k+5$. Now $12k+5 \pmod 8$: $12k \equiv 4k \pmod 8$, so $12k + 5 \equiv 4k + 5 \pmod 8$. For k odd: $4k \equiv 4 \pmod 8$, giving $12k+5 \equiv 1 \pmod 8$. For k even: $4k \equiv 0 \pmod 8$, giving $12k+5 \equiv 5 \pmod 8$. Since k is odd and even with equal density, $P[3][1] = P[3][5] = 1/2$ and $P[3][3] = P[3][7] = 0$.

Row 7 (restricted to $\{3, 7\}$). Write $n = 8k + 7$. Then $3n + 1 = 24k + 22 = 2(12k + 11)$. Since $12k + 11$ is always odd, $v_2(3n + 1) = 1$ exactly and $S(n) = 12k + 11$. Now $12k + 11 \pmod 8$: $12k + 11 \equiv 4k + 3 \pmod 8$. For k odd: $4k \equiv 4$, giving $4k + 3 \equiv 7 \pmod 8$. For k even: $4k \equiv 0$, giving $4k + 3 \equiv 3 \pmod 8$. Therefore $P[7][3] = P[7][7] = 1/2$ and $P[7][1] = P[7][5] = 0$. $\square$

Remark 2.3 (Forbidden transitions). The zero entries in P — transitions $3 \to 3$, $3 \to 7$, $7 \to 1$, $7 \to 5$ — are *structurally forbidden* for all n, not just on average. Classes 3 and 7 each map into a strict subset of $\{1, 3, 5, 7\}$. This structural restriction is the engine of the proof.

Remark 2.4 (Machine verification). All results in this theorem have been formally verified in Lean 4/Mathlib with no `sorry` or `admit` and no axioms beyond `propext`, `Classical.choice`, and `Quot.sound`. The key theorems are `S_class1/3/5/7` (arithmetic identities), `S_class1_mod8` (row 1 uniformity by direct mod-4 computation), `count_row1/3/7` (exact finite counts for rows 1, 3, 7), and `row5_count_not_exact` (counterexample for row 5 via `native_decide`: at $N = 6$, $b \equiv 1$ the count is 17, not 16). The Lean source is in `Paper67.lean` in the paper directory.

Remark 2.5 (The surviving chain has exact arithmetic). Row 5 of P is never used in the proof of Theorem 5.1. Within a sentence, class 5 is a sink: once the chain reaches 5 the sentence ends, and class 5 never acts as a source for subsequent steps. The surviving distribution (Definition 3.2) propagates mass only through rows 1, 3, and 7 of P.

For these three rows, all transition probabilities are *exact rational constants* that follow directly from $n \bmod 8$ alone: $v_2(3n + 1) = 2$ always for class 1, and $v_2(3n + 1) = 1$ always for classes 3 and 7. No geometric series, no expectation, no appeal to the distribution of $v_2(3n + 1)$ for class 5 is ever needed. The variable-valuation behaviour of class 5 — which requires an infinite geometric series to characterise (Paper 66 [3]) — is invisible to the surviving chain. This exactness is what makes the sentence decomposition the right scale for counting, and what makes the proof of Theorem 5.1 purely algebraic.

3 The Sentence as a First-Passage Problem

A sentence is the first-passage time of the Markov chain (P, π_0) to state 5, where π_0 is the sentence-start distribution.

Lemma 3.1 (Sentence-start distribution)**.** *The sentence-start distribution is* $\pi_0 = (1/4, 1/4, 1/4, 1/4)$ *— uniform on* $\{1, 3, 5, 7\}$.

Proof. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod 8$. By Theorem 2.1, row 5 of P is uniform, so b_0 mod 8 is uniform on $\{1, 3, 5, 7\}$. □

The sentence terminates at the first step where the chain visits state 5. The sentence length k is this first-passage time.

Definition 3.2 (Surviving distribution)**.** Let $d_s^{(k)}$ denote the probability that the chain is at state s after k steps without having visited state 5. Formally, $d_s^{(0)} = \pi_0(s)$ and for $k \geq 0$:

$$d_s^{(k+1)} \;=\; \sum_{t \neq 5} d_t^{(k)} P[t][s], \qquad s \neq 5.$$

The *surviving mass* is $M_k = \sum_{s \neq 5} d_s^{(k)}$.

The sentence length distribution follows from the surviving mass: M_{k-1} is the probability of surviving past step $k - 1$, and M_k is the probability of surviving past step k, so

$$P(\text{sentence length} = k) \;=\; M_{k-1} - M_k.$$

Expanding: $M_{k-1} - M_k \;=\; \sum_{s \neq 5} d_s^{(k-1)} - \sum_{s \neq 5} d_s^{(k)} \;=\; \sum_{t \neq 5} d_t^{(k-1)} \Big(1 - \sum_{s \neq 5} P[t][s]\Big) \;=\; \sum_{t \neq 5} d_t^{(k-1)} P[t][5]$.

4 The Invariant $d_3 = d_7$

The key structural fact is a symmetry between classes 3 and 7 in the surviving distribution.

Lemma 4.1 (Invariant)**.** $d_3^{(k)} = d_7^{(k)}$ *for all* $k \geq 0$.

Proof. By Definition 3.2, the surviving distribution sums only over sources $t \neq 5$; class 5 contributes nothing because once the chain reaches 5 the sentence terminates (the mass is absorbed, not propagated). The transition update for states 3 and 7 is therefore:

$$\begin{aligned} d_3^{(k+1)} &= d_1^{(k)} \cdot P[1][3] + d_3^{(k)} \cdot P[3][3] + d_7^{(k)} \cdot P[7][3] \\ &= \tfrac{1}{4}\, d_1^{(k)} + 0 \cdot d_3^{(k)} + \tfrac{1}{2}\, d_7^{(k)}, \\ d_7^{(k+1)} &= d_1^{(k)} \cdot P[1][7] + d_3^{(k)} \cdot P[3][7] + d_7^{(k)} \cdot P[7][7] \\ &= \tfrac{1}{4}\, d_1^{(k)} + 0 \cdot d_3^{(k)} + \tfrac{1}{2}\, d_7^{(k)}. \end{aligned}$$

The two expressions are identical: $d_3^{(k+1)} = d_7^{(k+1)}$ for any $d_1^{(k)}, d_3^{(k)}, d_7^{(k)}$. By induction, since $d_3^{(0)} = d_7^{(0)} = 1/4$ (Lemma 3.1), $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$. □

Remark 4.2 (Structural source of the invariant). The invariant holds because columns 3 and 7 of P, restricted to rows $\{1, 3, 7\}$, are *identical*:

$$P[\cdot][3]\big|_{\{1,3,7\}} \;=\; P[\cdot][7]\big|_{\{1,3,7\}} \;=\; \left(\tfrac{1}{4},\, 0,\, \tfrac{1}{2}\right).$$

This column symmetry is a direct consequence of Theorem 2.1:

- *From class 1*: uniform output gives equal weight 1/4 to both 3 and 7.
- *From class 3*: forbidden to reach 3 or 7 (both have weight 0).
- *From class 7*: reaches 3 and 7 with equal probability 1/2 each.

The invariant is unconditional — it holds for *any* value of d_1, not just at the fixed point of the chain.

5 Main Theorem

Theorem 5.1 (Sentence length distribution)**.** *For any $k \geq 1$:*

$$P(\textit{sentence length} = k) = \frac{3^{k-1}}{4^k}.$$

Equivalently, $P(\textit{sentence length} = k) = \frac{1}{3}\left(\frac{3}{4}\right)^k$.

Proof. We use the splitting: $P(\text{sentence length} = 1) = \pi_0(5) = 1/4$ (sentences that start at $b_0 \equiv 5$ terminate immediately). For $k \geq 2$, the sentence does not terminate at step 1 (probability 3/4) and then terminates at step $k-1$ from the conditional start, i.e. $P(\text{length} = k) = \frac{3}{4} \cdot g_{k-1}$, where g_j is the first-passage probability to 5 starting from the conditional distribution $\pi_0(\,\cdot \mid \cdot \neq 5) = (1/3, 1/3, 1/3)$ uniform on $\{1, 3, 7\}$.

It suffices to show $g_j = (3/4)^{j-1}/4$ for all $j \geq 1$, which follows if the surviving mass M_j starting from uniform $\{1, 3, 7\}$ satisfies $M_j = (3/4)^j$.

Step 1: $d_3 = d_7$ is preserved. The conditional start has $d_3^{(0)} = d_7^{(0)} = 1/3$, so Lemma 4.1 applies: $d_3^{(k)} = d_7^{(k)}$ for all $k \geq 0$.

Step 2: Surviving mass decays by exactly $3/4$. From Definition 3.2, $M_{k+1} = \sum_{s\neq 5} d_s^{(k+1)} = \sum_{s\neq 5}\sum_{t\neq 5} d_t^{(k)} P[t][s] = \sum_{t\neq 5} d_t^{(k)} \sum_{s\neq 5} P[t][s] = \sum_{t\neq 5} d_t^{(k)}(1 - P[t][5])$, giving:

$$M_{k+1} \;=\; d_1^{(k)}\big(1 - P[1][5]\big) + d_3^{(k)}\big(1 - P[3][5]\big) + d_7^{(k)}\big(1 - P[7][5]\big).$$

Substituting $P[1][5] = 1/4$, $P[3][5] = 1/2$, $P[7][5] = 0$:

$$M_{k+1} \;=\; \tfrac{3}{4}\, d_1^{(k)} + \tfrac{1}{2}\, d_3^{(k)} + d_7^{(k)}.$$

We claim $M_{k+1} = \frac{3}{4} M_k = \frac{3}{4}(d_1^{(k)} + d_3^{(k)} + d_7^{(k)})$. This requires:

$$\tfrac{3}{4}\, d_1^{(k)} + \tfrac{1}{2}\, d_3^{(k)} + d_7^{(k)} \;=\; \tfrac{3}{4}\, d_1^{(k)} + \tfrac{3}{4}\, d_3^{(k)} + \tfrac{3}{4}\, d_7^{(k)},$$

which simplifies to $\frac{1}{2}d_3^{(k)} + d_7^{(k)} = \frac{3}{4}(d_3^{(k)} + d_7^{(k)})$, i.e. $d_7^{(k)} = d_3^{(k)}$. This is exactly Lemma 4.1.

Step 3: Initial condition. At $k = 0$, $M_0 = 1$ (the full conditional mass on $\{1, 3, 7\}$). By induction, $M_k = (3/4)^k$ for all $k \geq 0$.

Conclusion. $g_j = M_{j-1} - M_j = (3/4)^{j-1} - (3/4)^j = (3/4)^{j-1}(1/4) = 3^{j-1}/4^j$. Therefore:

$$P(\text{length} = 1) \;=\; \tfrac{1}{4} \;=\; \tfrac{3^0}{4^1},$$

$$P(\text{length} = k) \;=\; \tfrac{3}{4} \cdot g_{k-1} \;=\; \tfrac{3}{4} \cdot \frac{3^{k-2}}{4^{k-1}} \;=\; \frac{3^{k-1}}{4^k}, \qquad k \geq 2.$$

Both cases give $3^{k-1}/4^k$, completing the proof. $\square$

Corollary 5.2 (Normalisation)**.** $\sum_{k=1}^{\infty} P(\textit{sentence length} = k) = 1$.

Proof. $\sum_{k=1}^{\infty} \frac{3^{k-1}}{4^k} = \frac{1}{4}\sum_{k=1}^{\infty}\left(\frac{3}{4}\right)^{k-1} = \frac{1}{4} \cdot \frac{1}{1-3/4} = 1.$ $\square$

Remark 5.3 (Machine verification of the Markov chain algebra). The Markov chain results — Lemma 4.1 (`invariant_d3_d7`), $M_k = (3/4)^k$ (`Mmass_eq`), $g_j = 3^{j-1}/4^j$ (`g_eq`), $P(\text{length} = k) = 3^{k-1}/4^k$ (`Plen_eq`), and the normalisation identity (`normalization`) — have all been formally verified in Lean 4/Mathlib with no `sorry`, no `admit`, and no axioms beyond the standard Lean kernel. The verification operates on exact rational arithmetic throughout; no real-analysis machinery is required. The Lean source is in `Paper67.lean` in the paper directory.

6 Discussion

6.1 The role of class 5

The formula $3^{k-1}/4^k$ is the geometric distribution with success probability 1/4, as if at each step the chain draws uniformly from $\{1, 3, 5, 7\}$ and terminates on drawing 5. This appearance is striking because class 5 plays three distinct and mutually reinforcing roles in the sentence structure.

Class 5 as the termination state. A sentence terminates exactly when it reaches a node $\equiv 5 \pmod 8$. Since class 5 is the unique absorbing boundary of the chain (once the chain visits class 5, the sentence ends), it is structurally absent from all intermediate steps. Empirically, class 5 accounts for only $\approx 6.2\%$ of within-sentence word entries (Appendix A, Table 2) — one entry per sentence, uniformly distributed across sentence lengths — rather than the 25% it would contribute if words were unrestricted.

Class 5 as the uniform source. The sentence starts at $b_0 = S(n)$ with $n \equiv 5 \pmod 8$. By row 5 of the transition matrix (Theorem 2.1), class 5 maps uniformly over $\{1, 3, 5, 7\}$: any of the four classes is equally likely as the first word. In particular, exactly 1/4 of sentences have $b_0 \equiv 5$, meaning length $k = 1$. This is the correct $P(k = 1) = 1/4$, not the naive 1/2. Class 5 is the unique class for which $v_2(3n + 1) \geq 3$ always (proved in Paper 66 [3]), which forces the extra halvings that spread the output uniformly.

Class 5 as the structural asymmetry. The transition $7 \to 5$ is structurally forbidden: $P[7][5] = 0$. Class 5 is therefore unreachable in one step from class 7, while it is reachable from classes 1 (with probability 1/4) and 3 (with probability 1/2). This asymmetry — class 5 reachable from some classes but not others — is precisely what forces the invariant $d_3 = d_7$: nodes in class 7 can only reach class 5 via a detour through class 3 or class 7, never directly. The proof in Section 4 shows that this forced detour maintains equal weight on classes 3 and 7 throughout the chain, and Section 5 shows that this equal weight is the algebraic condition that makes the effective termination probability exactly 1/4 at every step, regardless of the current value of d_1.

6.2 Connection to Paper 66

Paper 66 [3] proves that $E[v_2(3n+1) \mid n \equiv r \pmod 8]$ equals exactly 2, 1, 4, 1 for $r = 1, 3, 5, 7$ respectively. The two papers are complementary: Paper 66 establishes what happens *at* the class 5 boundary (a geometrically distributed valuation requiring an infinite series for its expectation), while the present paper shows that by choosing Steiner sentences as the unit of analysis, the class 5 boundary need never be crossed from the inside — it only needs to be *entered*.

More precisely, the 3/4 decay rate follows from three facts, each drawing on the same underlying arithmetic:

- The 1/4 probability of immediate termination ($b_0 \equiv 5$) uses row 5 of P exactly once — at the sentence start. Class 5 is the unique class with $c_r \geq 3$ (Paper 66), forcing the extra halvings that make its output uniform. This is the only point where the geometric-series character of class 5 is relevant.
- The 1/2 termination probability from class 3 comes from the mod-16 split of Theorem 2.1: $v_2(3n+1) = 1$ exactly for all $n \equiv 3 \pmod 8$, so $S(n) \bmod 8$ is determined by $n \bmod 16$ alone, with no random valuation.
- The invariant $d_3 = d_7$ — and hence the 3/4 decay — involves classes 1, 3, and 7 only, all of which have constant valuations. The proof is purely algebraic: no expectation, no series.

Paper 66 handles the hard case (class 5, variable valuation). Paper 67 sidesteps it by construction.

6.3 Why $3^{k-1}/4^k$ rather than $(1/2)^k$

The naive prediction $(1/2)^k$ arises from the "3-node coin flip" model: treat each word in the sentence as independently terminating whenever it visits the 3-node, with the 3-node split giving probability 1/2. Under this model only classes 3 and 7 can terminate, and if the within-sentence distribution were uniform over $\{1, 3, 5, 7\}$, the effective per-step termination probability would be:

$$\tfrac{1}{4} \cdot 0 \;+\; \tfrac{1}{4} \cdot \tfrac{1}{2} \;+\; \tfrac{1}{4} \cdot 1 \;+\; \tfrac{1}{4} \cdot \tfrac{1}{2} \;=\; \tfrac{1}{2},$$

where the terms correspond to classes 1 (never terminates via 3-node), 3 (terminates with prob 1/2), 5 (terminal by definition), and 7 (terminates with prob 1/2). This model is wrong in two ways. First, class 1 *can* reach class 5 directly: $P[1][5] = 1/4$, so the naive assignment of 0 to class 1 understates termination from class 1. Second and more importantly, class 7 *cannot* reach class 5 at all: $P[7][5] = 0$, so the naive assignment of 1/2 to class 7 overstates it. The net effect of these two corrections is a reduction in the effective termination rate from 1/2 to 1/4. The structural reason is the forbidden transition $7 \to 5$: a chain in class 7 must first transit to class 3 or 7 (remaining in the growth classes), where it may then reach class 5 on a subsequent step. The invariant $d_3 = d_7$ (Lemma 4.1) shows that these two errors cancel exactly and yield an effective rate of precisely 1/4, independent of d_1.

Acknowledgements

This paper is part of the speculative results stream (64+k) of the Collatz programme. The proof structure was found through numerical exploration in `notebooks/steiner-branch-model.ipynb`.

A Empirical Validation

This appendix records the complete numerical experiment establishing the sentence length distribution. The experiment was first reported in Paper 65 [2] and is reproduced here in full as the empirical record accompanying this paper's proof.

Sampling Protocol

1. Draw t uniformly at random from $[1, T_{\max}]$ with $T_{\max} = 10^{15}$.
2. Set $n = 8t + 5$ (so $n \equiv 5 \pmod 8$).
3. Compute $b_0 = S(n)$, the first Steiner word of the new sentence.
4. Follow consecutive Steiner circuits from b_0, counting words, until the first output $b \equiv 5 \pmod 8$. Record the word count as k.

$N = 10^5$ independent sentences were collected with seed 42. Independence between samples is ensured by the random draw of t; no orbit tails are shared.

Sentence-start Distribution

By Theorem 2.1, row 5 of P is uniform, so $b_0 = S(n) \bmod 8$ should be uniform on $\{1, 3, 5, 7\}$.

Within-sentence Steiner Word Entry Distribution

The entry class of each Steiner word within a sentence determines its termination behaviour. Class 5 words always terminate the sentence, so they appear only as the final word — their within-sentence frequency is structurally suppressed to approximately $1/(k+1)$ for a sentence of length k, averaging to $\approx 1/16$ empirically.

b_0 mod 8	Count	Observed fraction	Expected (uniform)
1	25181	0.25181	0.25000
3	25023	0.25023	0.25000
5	24852	0.24852	0.25000
7	24944	0.24944	0.25000

Table 1: Sentence-start distribution: $b_0 = S(n)$, $n \equiv 5 \pmod 8$, $N = 10^5$ samples. The distribution is uniform on $\{1, 3, 5, 7\}$ as predicted by Theorem 2.1.

Entry mod8	Count	Fraction	Visits 3-node?	Term. prob.
1	125450	0.31333	no	0
3	125119	0.31250	yes	1/2
5	24852	0.06207	no	1 (terminal only)
7	124955	0.31209	yes	1/2
Total	400376			

Table 2: Within-sentence Steiner word entry distribution across all 10^5 sentences (400376 total words). Class 5 appears only as the terminal word, giving frequency ≈6.2% rather than the uniform 25%. The 3-node fraction (classes 3+7) is 62.46%; the implied effective per-word termination probability is $0.6246/2 \approx 0.312$. The theoretical value is 1/4 (Theorem 5.1); Lemma 4.1 proves $d_3 = d_7$ which makes the exact rate 1/4.

Sentence Length Distribution

Statistical Tests

A χ^2 goodness-of-fit test on 15 length bins plus a tail bin, using $N = 10^5$ sentences:

Model	χ^2	p-value	Verdict
Naive $(1/2)^k$	1,022,029	≈ 0	**Rejected**
Theory $3^{k-1}/4^k$	12.9	0.61	Consistent

The naive model is rejected with overwhelming confidence. The theoretical formula $3^{k-1}/4^k$ is consistent with the data at all tested lengths.

References

[1] J. Seymour, *A Regular Expression Language for the Collatz Graph* (Working Paper, Paper 33), 2026.

[2] J. Seymour, *The $\frac{1}{3}(\frac{3}{4})^k$ Distribution of Steiner Sentence Lengths* (Working Paper, Paper 65), 2026.

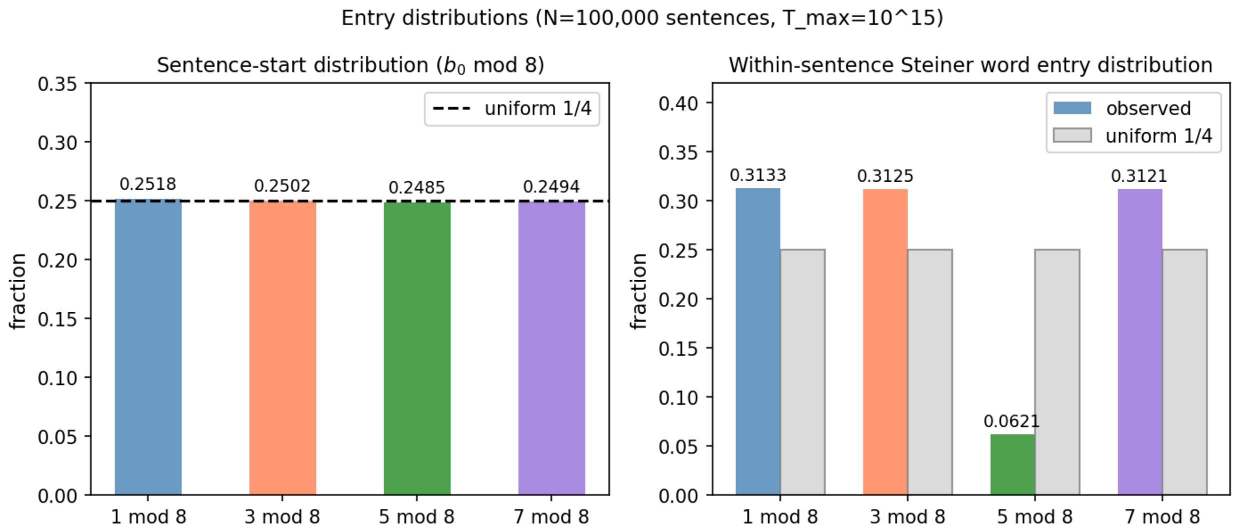

Figure 1: Left: sentence-start distribution (b_0 mod 8) against the uniform 1/4 baseline — confirmed uniform (Table 1). Right: within-sentence Steiner word entry distribution against the uniform 1/4 baseline. Class 5 is structurally suppressed to ≈6.2% (it can only appear as the terminal word); classes 1, 3, 7 each carry ≈31%.

[3] J. Seymour, *2-adic Valuations of* $3n + 1$ *by Residue Class mod 8* (Working Paper, Paper 66), 2026.

k	Observed fraction	Naive $(1/2)^k$	Theory $3^{k-1}/4^k$
1	0.25180	0.50000	0.25000
2	0.18683	0.25000	0.18750
3	0.13931	0.12500	0.14063
4	0.10574	0.06250	0.10547
5	0.07844	0.03125	0.07910
6	0.05958	0.01563	0.05933
7	0.04490	0.00781	0.04449
8	0.03342	0.00391	0.03337
9	0.02490	0.00195	0.02503
10	0.01866	0.00098	0.01877

Table 3: Empirical Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$) vs. the naive $(1/2)^k$ prediction and the theoretical $3^{k-1}/4^k$ (Theorem 5.1). The naive prediction overestimates $P(k = 1)$ by a factor of 2 and decays far too fast for $k \geq 3$.

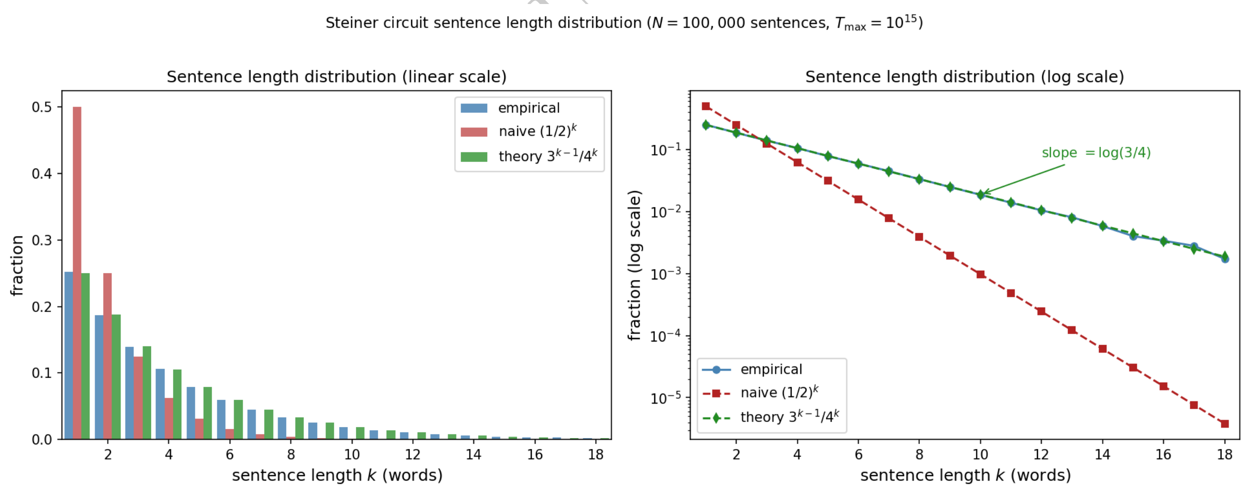

Figure 2: Steiner sentence length distribution ($N = 10^5$, $T_{\max} = 10^{15}$): empirical (blue bars / circles) vs. naive $(1/2)^k$ (red) and theoretical $3^{k-1}/4^k$ (green), on linear scale (left) and log scale (right). On the log scale, the empirical slope matches $\log(3/4)$ precisely — not $\log(1/2)$. The naive prediction is dramatically refuted at $k = 1$ (predicts 50%; observed 25%) and for $k \geq 3$ (predicts too-fast decay).